Homework
Eigenvalues and Eigenvectors

BCS A, E, F, G

Problem 1. Let $T(\boldsymbol{x}) = A\boldsymbol{x} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\boldsymbol{x}$ be the rotation through an angle of 90° in the counter clockwise direction. Find all eigenvectors and eigenvalues of A. Is there an eigenbasis for A?

Problem 2. Consider the reflection matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Find all eigenvectors and eigenvalues of A. By taking insights from the problem, what are the possible *real* eigenvalues of an orthogonal $n \times n$ matrix A?

Problem 3. For which $n \times n$ matrices A is 0 an eigenvalue? Give your answer in terms of the kernel of A and also in terms of the invertibility of A.

Problem 4. Find the eigenspaces of the matrix $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$. Give the geometrical interpretation.

Problem 5. Find the eigenspaces of

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Submission Date: September 28, 2024 (5:00PM)

Good Luck